

Project Initialization and Planning Phase

Date	July 9, 2026
Team ID	SWTID-2026-1262
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement):

Create a problem statement to understand your customer's point of view. The Customer Problem Statement helps you focus on what matters to create experiences people will love. A well-articulated customer problem statement allows you and your team to find the ideal solution for your customers' challenges. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1 (Farmer)	A local farmer managing changing soil quality and irregular seasonal weather patterns.	Maximize crop yield and allocate seed and fertilizer resources efficiently.	I lack access to real-time predictive data to accurately match soil chemistry (N,P,K) with upcoming environmental conditions.	I rely primarily on traditional guesswork and historical habits which are becoming less predictable.	Anxious and uncertain about crop survival, financial investment risks, and overall farm productivity.
PS-2 (Researcher)	An agricultural scientist analyzing crop-environment interactions.	Evaluate multi-model machine learning variations to model crop compatibility profiles accurately.	Gathering clean baseline datasets and evaluating multi-variable inputs (like humidity, temperature, and pH variance) is highly manual and fragmented.	Current analytical toolsets require complex code setups and lack consolidated web dashboards to verify target crop alignment.	Frustrated by delayed research progress and a lack of quick validation mechanisms for crop simulation models.